Youngjin (Luke) Kim

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Summary

PhD student in Electrical and Computer Engineering at USC specializing in EDA for advanced packaging. Experienced in physical design for **heterogeneous integration** (2.5D/3D ICs) with a focus on **signal and power integrity**, **GPU-accelerated EDA algorithm** optimization, and **ML-driven design automation**.

Education

University of Southern California (USC) — Los Angeles, CA
Ph.D. in Electrical and Computer Engineering — *Aug 2025 – Present*
- Selected Coursework: Hardware Foundations of ML, CAD of Digital Systems, Optimization for Information and Data Sciences

Yonsei University — Seoul, South Korea
B.S. in Electrical and Electronic Engineering – GPA: 3.83/4.0 (Major: 3.92/4.0) — *Mar 2019 – Aug 2025*

Skills

Programming & HDL: C/C++ (libtorch, CUDA, HLS), Python (PyTorch), Verilog, Chisel, MATLAB, Shell Script, Tcl
EDA Tools: Keysight (ADS), Synopsys (DC, PrimeTime, HSPICE), Cadence (Innovus, Virtuoso, Integrity 3D-IC, Voltus, Allegro, APD+, Sigrity, Clarity), Siemens (HyperLynx, Xpedition), Ansys (HFSS)
Tools: Git, Linux, Docker, LaTeX

Work Experience

PhD Intern, Machine Learning Architecture Systems — May 2026 – Dec 2026
Keysight Technologies, SAL (Software and AI Labs) – Supervisor: Dr. Fabio-Alessio Marino — *Calabasas, CA*
- Developed and shipped **ML models** (variants of GNNs, DeepONet, and Transformers) in **Keysight AI-Driven Designer**, using **C++/CUDA and libtorch** to predict **S-parameters within 0.6% error**.
- Optimized **ML training and inference** in Keysight's RL-based optimizer by automatically tuning batch sizes on shared GPUs, increasing **GPU memory utilization by 20%**.

Graduate Research Assistant — Aug 2025 – Present
University of Southern California, SCCAD Lab – Advisor: Prof. Sung-Kyu Lim — *Los Angeles, CA*
- Built OpenEYE, a differentiable framework for signal integrity optimization in 2.5D IC interconnects, using **custom CUDA kernels**, achieving **37× faster forward analysis** than a commercial tool.
- Building **DiSCO**, a framework for **signal and power integrity co-optimization** in 2.5D IC interconnects, optimizing chiplet PHY settings (TX/RX equalizers, drivers) under crosstalk and shared-PDN coupling.
- Developing MosaicGR, a **heterogeneous CPU–GPU routing engine** with **CPU-guided adaptive scheduling**.

Intern, Controller Development — Sep 2024 – Jan 2025
Samsung Memory Division, Signal Processing Task Group — *Hwaseong, South Korea*
- Designed a hardware-accelerated **vector search engine** for AI datacenter SSD controllers, achieving **20× speedup** over a CPU baseline across 1B+ data points
- Developed an **IVF+PQ pipeline** on FPGA using **C/C++-based HLS**, optimizing memory access and computational throughput with metadata-filtering prefetch mechanisms to reduce data pipeline latency

Undergraduate Research Assistant — Aug 2023 – Aug 2025
Yonsei University, HAI Lab & HSCS Lab – Advisors: Prof. Jaeyong Chung, Prof. Woo-Young Choi — *Seoul, South Korea*
- Designed a **RISC-V processor** and a Chisel-based reconfigurable computing array for OpenGeMM acceleration
- Developed FPGA-based digital control systems and thermal crosstalk cancellation algorithms for a **Silicon Micro Ring Modulator-based optical neural network**

Publications

DiSCO: Differentiable Framework for Signal–Power Integrity Co-Optimization of 2.5D Chiplet Links — 2027
- **Y. Kim**, M. G. Park, M. A. Dolatsara, R. Rabinovich, and S. K. Lim. *ACM/IEEE Design Automation Conference (DAC)*, 2027 (Work In Progress)

Invited: OpenEYE: A Differentiable Framework for Signal Integrity Optimization in 2.5D IC Interconnects — 2026
- **Y. Kim**, M. G. Park, M. A. Dolatsara, R. Rabinovich, and S. K. Lim. *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2026

A Simple and Efficient Thermal Crosstalk Cancellation Technique for Photonic Weight Banks in Optic Neural Networks — 2024
- **Y. Kim**, Y. Ji, and W. Choi. *OSK Summer Annual Meeting, Optics and Photonics Congress 2024*. Oral presentation, ICC Jeju, Korea, July 2024